

STATE TECHNICAL UNIVERSITY

END SEMESTER EXAMINATION 2025-2026 | ANSWER BOOKLET

CANDIDATE ROLL NO: 2026-CS-PH-017
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BARCODE SCAN TRACKING



BK-PH-80017

Q.No	Q1	Q2	Q3	Q4	Q5
Max	14	14	14	14	14
Marks					

START WRITING YOUR ANSWERS BELOW. DO NOT WRITE IN THE MARGINS.

Q.1 Ans 1. Quantum Physics: Group Velocity & Particle Velocity:

In quantum mechanics, a moving physical particle (mass m , momentum p) is associated with a wave packet rather than a single harmonic wave (de Broglie hypothesis: $\lambda = h/p$).

1. Phase Velocity (v_p):

Speed of propagation of individual phase crests of a monochromatic wave:

$$v_p = \omega / k = E / p = (p^2 / 2m) / p = p / (2m) = v / 2 \text{ (non-relativistic).}$$

2. Group Velocity (v_g):

Speed with which the overall modulation envelope (wave packet) moves:

$$v_g = d(\omega) / dk = dE / dp.$$

$$\text{Since } E = p^2 / (2m), dE/dp = 2p / (2m) = p / m = v \text{ (particle velocity).}$$

$$\text{Hence, } v_g = v_{\text{particle}}.$$

3. Relativistic Proof:

$$E^2 = p^2 c^2 + m_0^2 c^4.$$

$$\text{Differentiating: } 2E dE = 2p c^2 dp \Rightarrow dE/dp = (p c^2) / E.$$

$$\text{Substituting } p = \gamma m_0 v \text{ and } E = \gamma m_0 c^2:$$

$$v_g = (\gamma m_0 v c^2) / (\gamma m_0 c^2) = v.$$

Therefore, group velocity v_g is universally equal to classical particle velocity v .

Q.2 *Ans 2. Wave Optics - Interference of Light:*

Interference is the redistribution of luminous energy due to superposition of two coherent wave trains:

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\delta).$$

Conditions for Sustained Interference:

- 1. Sources must be strictly coherent (constant phase difference).*
- 2. Sources must be monochromatic (single wavelength λ).*
- 3. Equal or nearly equal amplitudes for maximum fringe contrast.*

Division of Wavefront vs Division of Amplitude:

a) Division of Wavefront:

- Original wavefront is divided into two parts by mirrors, prisms or slits.*
- Examples: Young's Double Slit, Fresnel Biprism, Lloyd's Mirror.*
- Requires a point or narrow slit source. Fringe width $\beta = \lambda \cdot D / d$.*

b) Division of Amplitude:

- Incident wave amplitude is split by partial reflection and refraction at interfaces.*
- Examples: Thin dielectric films, Newton's Rings, Michelson Interferometer.*
- Can use broad extended sources, producing high intensity fringes.*
- Path difference in thin film: $\Delta = 2 \cdot \mu \cdot t \cdot \cos(r) - \lambda / 2$ (Stokes' phase shift).*
- Applications: Anti-reflection optical coatings ($t = \lambda / 4\mu$) and optical surface testing.*

Q.3**Ans 3. Lasers and Fibre Optics:****1. Laser Principles (Light Amplification by Stimulated Emission of Radiation):**

- Stimulated Absorption: Ground atom absorbs $h\nu$ and jumps to E_2 .
- Spontaneous Emission: Random downward transition emitting photon: Rate = $A_{21} * N_2$.
- Stimulated Emission: Incident photon induces excited atom to emit an identical twin photon (same frequency, phase, direction): Rate = $B_{21} * N_2 * \rho(\nu)$.
- Population Inversion ($N_2 > N_1$): Non-equilibrium state achieved via optical/electrical pumping in a 3-level or 4-level system with a metastable state (lifetime $\sim 10^{-3}$ s).
- Optical Resonator: Pair of mirrors (100% and 95% reflection) creating standing waves: $L = m * \lambda / 2$.

2. Fibre Optics Fundamentals:

- Wave propagation occurs by Total Internal Reflection (TIR) at Core-Cladding interface ($n_1 > n_2$).
- Critical Angle: $\theta_c = \sin^{-1}(n_2 / n_1)$.
- Acceptance Angle (θ_0): Maximum entry angle:
 $\sin(\theta_0) = \sqrt{n_1^2 - n_2^2} = NA$ (Numerical Aperture).
- Fractional Refractive Index Change: $\Delta = (n_1 - n_2) / n_1$.
- $NA = n_1 * \sqrt{2 * \Delta}$.
- Types: Step Index (Single mode / Multimode) and Graded Index (GRIN) fibers with parabolic index profile reducing intermodal dispersion.

Q.4**Ans 4. Electromagnetism - Maxwell's Equations & Boundary Conditions:****1. Maxwell's Four Fundamental Equations:****a) Gauss's Law for Electrostatics:**

$$\text{Differential: } \text{div}(\mathbf{D}) = \rho_v \quad | \quad \text{Integral: } \oiint \mathbf{D} \cdot d\mathbf{A} = Q_{\text{enclosed}}$$

Physical meaning: Electric flux through closed surface equals total enclosed charge.

b) Gauss's Law for Magnetism:

$$\text{Differential: } \text{div}(\mathbf{B}) = 0 \quad | \quad \text{Integral: } \oiint \mathbf{B} \cdot d\mathbf{A} = 0$$

Physical meaning: Magnetic monopoles do not exist; magnetic lines form continuous loops.

c) Faraday's Law of Electromagnetic Induction:

$$\text{Differential: } \text{curl}(\mathbf{E}) = -d\mathbf{B}/dt \quad | \quad \text{Integral: } \oint \mathbf{E} \cdot d\mathbf{l} = -d(\Phi_B)/dt$$

Physical meaning: Time-varying magnetic fields induce circulating electric fields.

d) Ampere-Maxwell Law:

$$\text{Differential: } \text{curl}(\mathbf{H}) = \mathbf{J} + d\mathbf{D}/dt \quad | \quad \text{Integral: } \oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{cond}} + d(\Phi_E)/dt$$

Displacement current density: $\mathbf{J}_d = d\mathbf{D}/dt = \epsilon \cdot d\mathbf{E}/dt$, resolving continuity equation $\text{div}(\mathbf{J}) + d\rho/dt = 0$.

2. Boundary Conditions at Dielectric Interfaces:

- Tangential E-field is continuous: $E_{1_t} = E_{2_t}$.
- Normal D-field discontinuity: $D_{1_n} - D_{2_n} = \rho_s$ (if $\rho_s = 0$, $D_{1_n} = D_{2_n}$).
- Tangential H-field continuity (for non-conducting boundary): $H_{1_t} = H_{2_t}$.
- Normal B-field is always continuous: $B_{1_n} = B_{2_n}$.

3. Poynting Vector: $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ (W/m^2), representing instantaneous directional power flow per unit area.

Q.5 Ans 5. Solid State & Semiconductor Physics:

1. Crystal Structures and Lattice Geometry:

- Simple Cubic (SC): Atoms at 8 corners, Coordination No (CN) = 6, Atomic Packing Factor (APF) = 52%.
- Body Centered Cubic (BCC): Atoms at corners + body center, CN = 8, APF = 68%.
- Face Centered Cubic (FCC): Atoms at corners + face centers, CN = 12, APF = 74%.
- Miller Indices (h k l): Reciprocal of planar intercepts cleared of fractions. Interplanar distance: $d = a / \sqrt{h^2 + k^2 + l^2}$.

2. Energy Band Theory:

- Kronig-Penney Model explains periodic lattice potential causing splitting of energy levels into allowed bands and forbidden gaps (E_g).
- Conductors ($E_g = 0$ eV), Semiconductors ($E_g \sim 1.1$ eV for Si), Insulators ($E_g > 5$ eV).

3. Intrinsic vs Extrinsic Semiconductors:

- Intrinsic (pure Si): $n_i = p_i = \sqrt{N_c \cdot N_v} \cdot \exp(-E_g / 2kT)$. Fermi level E_f lies in middle of bandgap.
- n-type: Doped with pentavalent donors (P, As); majority carriers electrons; E_f shifts close to conduction band.
- p-type: Doped with trivalent acceptors (B, Al); majority carriers holes; E_f shifts close to valence band.

4. Direct vs Indirect Bandgap:

- Direct (e.g. GaAs): Conduction band minimum & valence band maximum align at $k = 0$ ($\Delta k = 0$); efficient radiative recombination (ideal for LEDs/lasers).
- Indirect (e.g. Si): Peak and valley are misaligned; requires phonon participation, releasing heat.

--- End of Answer Booklet (All 5 questions attempted) ---

Candidate Sign: Diya